

# The Operator's Manual

*How to run, configure and operate the BSS — by role and by task. The companion volume, *The Honest Machine* (a build memoir · verify everything — how one person and an AI built a complete telecom suite and proved every piece of it), tells how it was built and why; this book tells you which button, which endpoint, which environment variable. Everything here is proven by the suite named beside it.*

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## 1 • Getting started

```
docker compose up -d          # ~90 containers; first build takes a while
open http://localhost:8080/shop/ # the storefront (guest browsing works)
```

The gateway is :8080 ; Keycloak is :8085 (admin/admin). Two seed operators run side by side: **genalpha** (EN/EUR, realm `bss` ) and **nova** (NO/NOK, realm `nova` ) — plus any operator you

mint (§7). Demo card 4242 4242 4242 4242 pays; anything ending 0002 declines. Promo code WELCOME10 . Serviceable fiber postcodes start 111 / 222 / 333 .

**The one discipline:** every feature in this manual has an end-to-end Playwright suite in `ops/e2e/` . When in doubt about how something behaves, the suite is the specification: `node ops/e2e/<name>_test.js` runs it against the live stack.

## 2 • Surfaces & sign-ins

| URL                                         | WHO                                      | DEMO SIGN-IN                                                                           |
|---------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------|
| <code>/shop/</code>                         | Customers (self-register or guest)       | <code>kai@bss.local</code> / <code>kai</code>                                          |
| <code>/app/</code>                          | Customers on mobile (Expo web)           | <code>emil@acme.example</code> / <code>emil</code>                                     |
| <code>/biz/</code>                          | B2B org admins + invited members         | <code>bianca@acme.example</code> / <code>bianca</code>                                 |
| <code>/csr/</code>                          | Agents (role-scoped powers)              | <code>agent-anna</code> / <code>agent ; junior: jo@bss.local</code> / <code>jo</code>  |
| <code>/console/</code>                      | Back office (role-scoped tabs)           | <code>demo</code> / <code>demo ; product-only: pat@bss.local</code> / <code>pat</code> |
| <code>/dealer-app/</code>                   | Retail-partner clerks & telesales agents | any account whose org holds a dealer agreement                                         |
| <code>shop.nova.localhost:8080/shop/</code> | Nova's white-label storefront            | <code>nils@nova.local</code> / <code>nils</code>                                       |
| <code>biz.nova.localhost:8080/biz/</code>   | Nova's B2B console (Norwegian)           | <code>birgit@fjellheim.no</code> / <code>birgit</code>                                 |

White-labeling is hostname-driven: `shop.<tenant>.localhost` serves that operator's brand, locale and currency from the live registry (§7).

## 3 • The product owner

**Catalog (console → Product Offerings / Specifications / Prices)**

Offerings are TMF620 data: hard and soft bundles ( `bundledProductOfferingOption` cardinality, pick-N-of-M choice groups), characteristic-conditioned prices ("+2.00/mo when colour = Titanium"), commitment terms, categories that drive placement *and* fulfilment (Top-ups and Insurance are billing-only; Partner services mint entitlement codes; everything else draws an MSISDN + SIM). Artwork rides the offering's attachments — internal store or the tenant's own PIM; the internal store's bytes live in-row by default or in S3-protocol / Azure Blob object storage (deployment config, §9 — suite #54 proves all three).

## Rules & pricing as data (console → Rules)

Order rules and dynamic pricing are JSON-logic rows with a dry-run panel. A price change is a rule, not a release. See `docs/product-rules.md` for the authoring guide.

## Product Copilot (console → Copilot)

Chat a product into the catalog: the model proposes specs/prices/offerings as TMF620 payloads, the console shows a human card, and a deterministic executor applies it with *your* token on confirmation. The model never writes. The copilot also authors **pricing rules** ("10% off the phone with this plan") and — suite #61 — **personalization experience rules** ("device browsers see this banner and this pin"): both become policy rows you can disable in Rules. **Voice input:** the  button uses the browser's speech recognizer (Chrome/Edge; hidden where unsupported) — the transcript lands in the input and the human still presses Send; the mic never auto-sends. Server-side STT (bring-your-own Whisper) is a named seam for later.

## Product Advisor (console → Product advisor) — suite #52

Findings with receipts: top-up attach counted from inventory ("2 of 2 customers on this plan also bought top-ups"), market price gaps from the tenant's market feed (10% materiality threshold — small gaps are deliberately silent), an LLM narrative that never invents a number. **Adopt as draft...** births an `In study` offering; nothing reaches the shop until you promote it.

## Campaigns & growth (console → Campaigns / Journeys / Audiences)

Segments form from consented shop behavior; campaigns prove lift against real holdouts; journeys read the customer before speaking; A/B arms split deterministically; guardrails (frequency caps, quiet hours) are settings. Every message lands on the TMF683 timeline.

## 4 • The CSR

Search the 360 (multi-word, name-ranked, with a trigram typo net — "Solvieg" finds Solveig when strict matching finds nothing), then act — every action logs itself on the customer's timeline:

| THE CUSTOMER ASKS                 | THE BUTTON / ACT                                                                                |
|-----------------------------------|-------------------------------------------------------------------------------------------------|
| "What's my PUK?" / "reset my PIN" | Reveal PUK · Reset PIN (owner notified — a silent credential change reads as fraud)             |
| "I lost my SIM"                   | Replace SIM (old card quarantined first)                                                        |
| "new number, please"              | Change number (old number quarantined with its story)                                           |
| "pause my subscription"           | Pause / Resume (charging pauses at the OCS; auto-resumes on the promised date)                  |
| "I can't pay all of it"           | Installments (parts sum exactly; miss one → reminder → grace → dunning)                         |
| "this charge is wrong"            | Dispute (collection pauses; resolution credits or upholds, in writing)                          |
| "send me my invoice"              | PDF (the exact document) · Email bill (to the address on file, never one dictated on the phone) |
| "my internet is slow"             | Diagnose (triage across usage/assurance/network facts)                                          |
| "give my number to my colleague"  | Transfer (three-way authorization; B2B handover keeps the company paying)                       |

Self-care mirrors most of these in the shop and app — customers act on their OWN line (ownership-checked; a foreign line answers 404, never 403).

### The knowledge base at the desk

Articles are audience-shelved (customer / CSR / sales / product owner) and the CSR's ask-with-sources grounds its answers in them. Search is full-text, ranked, and stems in **the tenant's own language** — the locale that picks the storefront's currency also picks the stemmer, so on nova "regning" finds "regningene" (suite #55). Typos in customer search are caught by a trigram net that only speaks when strict matching finds nothing — strict results are byte-identical to before. Beneath the keyword search sits the **semantic net** (pgvector): "why is my internet so slow" finds the fair-use article that contains neither word, and nonsense finds nothing — the cosine ceiling keeps the net honest. Embeddings are a seam: `bss.knowledge.embeddings = stub` (deterministic, keyless) or `openai-compatible` (any real model).

## 5 • Billing & money operations

### The billing run

`POST /tmf-api/customerBillManagement/v4/billingRun` (staff). Segment-based and per-account: mid-month starts, mid-cycle plan changes and ceases each bill exactly their own days, labelled ("from the 16th, 15 days"). Payday alignment per customer ( `POST /individual/{id}/billingCycle` , day 1–28). The invariant the suites enforce from every direction: **no day is ever billed twice**.

### The bill as a document — suite #46

`GET /customerBill/{id}/document.pdf` (owner-scoped); `POST /customerBill/{id}/resend` emails it to the address on file. Delivery preference per customer: `paper | einvoice | digital` ( `POST /individual/{id}/billDelivery` ) — the customer picks the channel; the tenant keeps the plumbing.

### E-invoice formats are rows (console → Bill formats)

| CODE                 | STANDARD                                  | SYNTAX             |
|----------------------|-------------------------------------------|--------------------|
| <code>ehf</code>     | EHF Billing 3.0 (Norway; carries the KID) | UBL 2.1            |
| <code>peppol</code>  | Peppol BIS Billing 3.0                    | UBL 2.1            |
| <code>cii</code>     | EN 16931                                  | UN/CEFACT CII      |
| <code>aunz</code>    | A-NZ Peppol BIS 3.0                       | UBL 2.1            |
| <code>edifact</code> | UN/EDIFACT INVOIC D96A                    | segments           |
| <code>facturx</code> | Factur-X / ZUGFeRD                        | PDF + embedded CII |

Adding a country is an INSERT (the suite added XRechnung through the form). The tenant's `BILL_DISTRIBUTION_FORMAT` picks a row by code; the renderer follows the row.

### Delivery, status, and the money coming home — suites #46/#47

**Deliveries** (console tab): the outbox-backed ledger — every bill's trip, attempts, the buyer's Peppol Invoice Response (accepted / rejected with their words / paid), one-click Retry on failures. **Remittance**: the bank posts camt.054, Nets OCR or BAI2 to `POST /bank/v1/remittance` (per-tenant `BANK_TOKEN` ); a matched KID settles the bill through the

card path's own guarantee; everything unclear parks on **Unapplied cash** (console tab) with its reason. Money is fail-closed: never guessed, never dropped.

## Dunning (console → Dunning)

The collections worklist: overdue installments, broken plans, what is still owed.

## 6 • Partner channels — suites #48/#51

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**Being a dealer is a row.** Sign a chain: `POST /dealer/v1/agreement` (staff) with the org, commission per activation, and — for a chain with its own POS — the `OAuth2 clientId` that speaks for them. Clerks are org membership; foreign dealers see 404-shaped nothing.

### Retail (the CSP + external-retail model — think Elkjøp/Power)

**Counter sale:** sell against the customer's email; the order carries the dealer stamp. **Starter kits:** mint batches (code + SIM + attribution in the box); the customer self-activates ( `POST /dealer/v1/starterKit/activate` ) and the line rides the SIM from the box. **POS API:** same endpoints, machine credential, per-partner rate limits at the edge (429 + Retry-After); the chain's own phone rides the sale as `device` context — never a billable item. **Commission:** pending → earned after the angreter window → clawed back with the reason if the customer leaves inside it.

### Telesales (outbound, shaped by the law)

**Dial list:** `GET /dealer/v1/telesales/dialList?segment=` — consented at the source (insight segments), every number DNC-washed, reserved citizens excluded *and counted*. **The call makes an OFFER** ( `POST /dealer/v1/telesales/offer` ): washed fail-closed (reserved refuses, unreachable register refuses, unconfigured tenant refuses); no order exists yet. **The customer confirms in writing** ( `POST /telesales/v1/confirm` , signed in, code from their inbox — or, for a cold prospect, after registering with the offered email); only then the order, the activation and the commission are born. Silence expires on a clock. Every dialer call logs to TMF683.

## 7 • The host administrator

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### The tenant fleet is a file

`infra/tenants/tenants.yml` is the shared registry (mounted into all 31 services); every service live-refreshes it. NEW operators join the running fleet within one refresh interval; changes to a serving operator's seams/brand/hosts apply live; identity fields ( `id` , `issuer` , key endpoints) require a restart — by design.

## Minting an operator — suites #49/#50

```
# the script (~2 minutes to a billed customer):
ops/onboard-tenant.sh fjord "Fjord Mobil" da DKK "#1B6CA8"

# or the form: console → Operators → five fields → Save
#   (host-tenant admins only; realm clone + registry block + seeded catalog;
#   shop.<id>.localhost wears the brand the moment the gateway refreshes)
# live rebrand of a form-born operator:
PATCH /onboarding/v1/operator/{id} {"name","color","locale","currency"}
```

Seed operators (genalpha, nova) are protected from the form — their config is env.

## Staff & roles (console → Staff)

Grant/revoke whole areas per operator over the tenant's own IdP (TMF672) — no Keycloak console needed for daily administration.

# 8 • AI configuration & governance — suites #53/#59

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## The individualized shop (suite #60)

`GET /ai/v1/forYou` — the signed-in customer's own rail (self-scoped: party = token subject, no parameter to probe). TMF680 candidates + consented interests + churn-risk loyalty flag; ONE governed FAST caption per 5-minute cache window ("Picked for you after your look at devices"). No personalization consent → zero interests, generic caption. The storefront renders it as the "Recommended for you" shelf, and the mobile app's For-you card rides the same endpoint (same cache — the second channel is free).

## The control plane (suite #59)

Every LLM turn is **metered** (token counts + cost land on the AI Audit ledger — the console's AI Audit tab shows tokens, cost and outcome per turn) and **governed** per tenant: `GET /ai/v1/governance` shows spend this window, the ceiling, remaining headroom and the **agent-action trail** (which AI wrote which resource — e.g. the advisor's

`catalog.createDraftOffering` ). `POST /ai/v1/governance/budget {"budgetMicros": 5000000, "windowHours": 720, "enabled": true}` sets the ceiling and the **kill-switch**; crossing the ceiling refuses fail-closed (429, audited as `refused-budget` ), `enabled=false` refuses instantly (403). No budget row = unlimited-and-enabled — governance is opt-in per tenant. Cost is estimated (`chars/4 tokens × per-model rate, bss.ai.prices.<model>` in micros per 1K, default `bss.ai.default-price-per-1k-micros` ); exact provider usage is a wired-later seam.

Per tenant, all data: `ai-provider` ( `stub` — keyless CI default — `openai-compatible` , `anthropic` ), `ai-base-url` , `ai-api-key` , `ai-model` . **Tiered routing**: the task class lives at the call site — FAST (copywriting, summaries, narratives) vs SMART (product proposals, next-best-offer; the default when unclassified). Every value resolves tier-first:

```
AI_MODEL_FAST=swift-mini           # cheap model for volume work
AI_MODEL_SMART=frontier-x          # careful model for judgment
AI_PROVIDER_SMART=anthropic        # tiers go down to the PROVIDER:
AI_BASE_URL_SMART=https://api...   # a local endpoint for volume,
AI_API_KEY_SMART=...               # a frontier API for judgment — at once
```

## 9 • The seam catalog

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Every external system is a per-tenant seam: configure it and the feature lights up; leave it blank and the feature degrades honestly (fail-open for enrichment, fail-closed for money, identity and consent).



| SEAM                             | TENANT FIELDS (ENV)                                                               | BEHAVIOR WHEN BLANK                                                                                 |
|----------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Email provider (ESP)             | DELIVERY_PROVIDER, ESP_URL, ESP_API_KEY, ESP_FROM                                 | in-app inbox only                                                                                   |
| Bill distribution partner        | BILL_DISTRIBUTION_{PROVIDER,URL,TOKEN,FORMAT,CHANNEL}                             | bills stay in-app/PDF                                                                               |
| Bank (remittance)                | BANK_TOKEN                                                                        | webhook refuses (money: fail-closed)                                                                |
| Do-not-call register             | DNC_URL, DNC_TOKEN                                                                | outbound telesales refuses (opt-IN by configuring)                                                  |
| Market-intelligence feed         | MARKET_FEED_URL, MARKET_FEED_TOKEN                                                | advisor's market findings absent; own-data findings stand                                           |
| LLM                              | AI_* (§8)                                                                         | keyless stub answers                                                                                |
| PIM (product imagery)            | PIM_BASE_URL                                                                      | internal TMF667 store                                                                               |
| Content store (deployment-level) | CONTENT_STORE = in-row   s3   azure-blob (+ S3_* / AZURE_* )                      | bytes in the document row (dev-simple)                                                              |
| Embeddings (deployment-level)    | bss.knowledge.embeddings = stub   openai-compatible (+ embeddings-url/key/model ) | deterministic keyless stub — the semantic net still works, with curated synonyms instead of a model |
| Web analytics (GA4)              | ANALYTICS_*                                                                       | internal insight only                                                                               |

|                 |                                     |                                |
|-----------------|-------------------------------------|--------------------------------|
| Social platform | SOCIAL_API_URL, SOCIAL_ACCESS_TOKEN | no audience push / lead import |
|-----------------|-------------------------------------|--------------------------------|

Per-tenant variants ride suffixes: `_NOVA`, `_FJORD`, ... matching the placeholders in `tenants.yml`.

## 10 • Environment reference (dev clocks)

| VARIABLE                                             | MEANING                                                        | DEV / PROD DEFAULT     |
|------------------------------------------------------|----------------------------------------------------------------|------------------------|
| BSS_BILLING_DUNNING_TICK_MS /<br>_GRACE_MS           | collections sweep / grace                                      | 5s & 20s / 60s & 7d    |
| BSS_BILLING_DISTRIBUTION_TICK_MS /<br>_RETRY_SECONDS | delivery relay / backoff base                                  | 3s & 4s / 30s & 60s    |
| BSS_SOM_COMMISSION_WINDOW_MS /<br>_TICK_MS           | angrerett window / harden tick                                 | 15s & 5s / 14d & 1h    |
| BSS_SOM_TELESALES_EXPIRY_MS /<br>_TICK_MS            | offer expiry / sweep                                           | 15s & 5s / 48h & 1h    |
| BSS_TENANTS_REFRESH_MS                               | live registry refresh                                          | 15s                    |
| BSS_GATEWAY_PARTNER_RATE_CAPACITY /<br>_WINDOW_MS    | per-partner rate limit (dealer surface)                        | 10 per 2s / 60 per 60s |
| GLOBAL_RATE_CAPACITY /<br>GLOBAL_RATE_WINDOW_MS      | the wide ring — a ceiling on EVERY path, per subject/client/IP | 1200 per 60s           |

## 11 • Extension API reference (beyond the TMF standard surfaces)

| ENDPOINT                                                                   | WHO                                  | DOES                                     |
|----------------------------------------------------------------------------|--------------------------------------|------------------------------------------|
| GET /customerBill/{id}/document.pdf                                        | owner/staff                          | the bill as a PDF                        |
| POST /customerBill/{id}/resend                                             | owner/staff                          | email the invoice to the address on file |
| POST /customerBill/{id}/dispute · /installmentPlan                         | owner/agent                          | contest / split a bill                   |
| GET/PATCH/POST /billFormatProfile[/code]]                                  | read: billing · write: billing:admin | e-invoice profiles as rows               |
| GET /billDistribution · POST /billDistribution/{id}/retry                  | billing:admin                        | the delivery ledger                      |
| POST /bank/v1/remittance                                                   | the bank (X-Bank-Token)              | camt.054 / OCR / BAI2 in                 |
| POST /distribution/v1/response                                             | the partner (X-Distribution-Token)   | Peppol Invoice Response onto the ledger  |
| GET /remittance/unapplied                                                  | billing:admin                        | unapplied cash worklist                  |
| POST /individual/{id}/billingCycle · /billDelivery                         | self/staff                           | payday alignment · delivery preference   |
| /dealer/v1/* (agreement, kits, sell, commission, orders/{id}, telesales/*) | staff / dealer / partner POS         | the whole dealer & telesales channel     |
| POST /telesales/v1/confirm                                                 | the customer, signed in              | the written yes that births the order    |
| GET /advisor/v1/findings · POST /advisor/v1/adopt                          | catalog:write                        | product advisor · adopt-as-draft         |
| GET/POST /onboarding/v1/operator · PATCH .../{id}                          | host roles:admin                     | mint / live-mutate an operator           |
| GET /app/tenant-config.json                                                | public, by Host                      | the white-label manifest                 |

## 12 • The mock integrations (dev stand-ins)

| CONTAINER         | PORT  | STANDS IN FOR                                                       | CHAOS / INSPECTION                                             |
|-------------------|-------|---------------------------------------------------------------------|----------------------------------------------------------------|
| mock-esp          | 8121  | SendGrid-shaped email provider                                      | GET /mails , webhook receipts                                  |
| mock-social       | 8122  | Meta-shaped audiences/leads                                         | —                                                              |
| mock-distribution | 8124  | Peppol access point / print house (validates EHF/EDIFACT/Factur-X!) | GET /invoices , POST /outage , POST /invoices/{billNo}/respond |
| mock-dnc          | 8125  | reservation register                                                | numbers ending 9999 are reserved; POST /outage                 |
| mock-market       | 8126  | tariff-comparison feed                                              | GET /offers                                                    |
| minio             | 9000  | S3-protocol object store (AWS/R2/on-prem)                           | —                                                              |
| azurite           | 10000 | Azure Blob Storage (Microsoft's emulator)                           | —                                                              |
| mock-llm          | 8127  | OpenAI- and Anthropic-dialect LLM                                   | names the model in answers; GET /requests                      |

## 13 • Production operations — suites #56/#57

### Ticks that never double-fire

Every mutating scheduled job — dunning, bill delivery, campaign journeys, churn sweeps, porting cutover, commission hardening, order resume, telesales expiry, cart abandonment — claims a row-lease ( `tick_lock` table in the service's own database) before it runs. Scale any service to N replicas and each tick still fires once; a crashed holder's lease expires on its own. Inspect a service's leases with `psql -d billing -c "SELECT * FROM tick_lock"`. Suite #56 proves it by running TWO billing replicas and counting exactly one delivered bill at the receiving partner.

### The billing run — partitioned, resumable, and on a ledger

Each account's bill commits in its *own* transaction; a failed account is recorded and skipped, never the run. A crashed run RESUMES on the next trigger — every bill already cut is its own checkpoint — and every run is a row: `GET /tmf-api/customerBillManagement/v4/billingRun`

lists recent runs (status running/completed/superseded, accounts, bills, failures, last error). One run speaks at a time: a trigger during a live run answers `{"busy": true}` (the lease heartbeats per account, so a crashed run frees it within ~2 minutes), and the database enforces one bill per account per period with a unique index — constraint, not convention. Suite #57 kills billing mid-run and proves exactly-one-bill-per-customer after the resume.

`BSS_BILLING_RUN_ACCOUNT_DELAY_MS` (default 0) is the dev pacing knob the suite uses to hold a run open.

## Load numbers and the smoke SLO

`node ops/load/loadtest.js --seconds 15 --workers 10` — plain-Node closed-loop driver, three scenarios (anonymous catalog, storefront page, authenticated bill reads), reporting req/s and p50/p95/p99. LIFT THE WIDE RING FIRST ( `GLOBAL_RATE_CAPACITY=1000000` `docker compose up -d gateway` ) or the Po ceiling will throttle the harness — that is it working. Baselines and their caveats: [docs/perf-baselines.md](https://docs.gene.com/perf-baselines.md).

## Alerts

`infra/prometheus/alert-rules.yml` : ServiceDown (2 min unscraped), OutboxPublishFailures (events queuing, not lost), GatewayServerErrors (5xx > 5%). The whole fleet is scraped (32 targets). View at `:9090/alerts` ; a deployment routes them to a pager via Alertmanager.

## Backup and the restore drill

`ops/backup.sh` dumps every database and role to `backups/` (git-ignored, newest 14 kept). `ops/restore-drill.sh` restores the newest dump into a throwaway container, verifies databases and rows, optionally proves a sentinel ( `--expect <db> "<sql>"` ), and removes the container. Run the drill on a schedule — an untested backup is a hope, not a backup.

## Rate limits, two rings — buckets in Redis

The dealer surface keeps strict per-partner buckets; every other path passes a generous fleet-wide ceiling (per subject for people, per client for machines, per IP for anonymous), answering `429 + Retry-After` when tripped. Dials in \$10. The buckets live in Redis ( `BSS_GATEWAY_REDIS_URL` ; blank = in-memory): replicas share ONE exact ceiling, a restarted gateway still refuses inside the same window, and an unreachable Redis fails open — fairness, never authorization.

## The live k8s soak (on demand) — local, AWS, and Azure

The Helm chart has run on three live clusters: local k3s, AWS EKS, and Azure AKS — each with billing at 2 replicas holding ONE set of tick leases, requests served through the in-cluster

gateway. The cloud runs are one command each: `ops/k8s-soak/eks-run.sh up|smoke|down` and `ops/k8s-soak/aks-run.sh up|smoke|down` (compose must stop first — one laptop, two worlds). Both wrap Terraform + registry push + managed-Postgres + Helm + smoke + teardown. The cloud-by-cloud differences (all absorbed by seams — the app never changed) are tabulated in [architecture.md §5](#); the receipts and the live-run truths each cloud taught are in [k8s-soak-plan.md](#). Smoke: `ops/k8s-soak/smoke.js` against port-forwarded gateway/keycloak.

**Managed-Postgres note:** point `config.dbHost` at the managed server and set `local.postgres.enabled=false`. Some tiers cap connections low (Azure B1ms  $\approx$  50) — the chart now holds only one idle connection per pool; raise the server's `max_connections` for headroom. Managed Postgres may also gate extensions (Azure needs `pg_trgm/vector` allow-listed) and demand TLS (add `sslmode=require` to the datasource URL, or relax on the server for a soak).

## The secret gate

`ops/install-hooks.sh` wires `ops/scan-secrets.sh` as a pre-commit hook: any staged diff carrying a real-secret shape (Anthropic/OpenAI/AWS/GitHub/Slack keys, private-key blocks) refuses to commit. Injectable values are listed in `.env.example`; what a real deployment must add beyond this stack — external secrets, TLS in transit, managed HA Postgres/Kafka — is the runbook, [docs/hardening.md](#).

## 14 • Privacy & GDPR operations — suite #58

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### The data passport (right of access)

`GET /privacy/v1/export` — a customer exports THEMSELVES with their own sign-in (no role needed; the fan-out rides their token to every service). The DPO — any user with `roles:admin` — adds `?partyId=` to export another person. The JSON names its categories and also the legal-hold categories it cannot delete.

### Erasure (right to be forgotten)

`POST /privacy/v1/erase {"partyId": "..."} (DPO only)`. Refuses 409 while the person holds ACTIVE services — terminate first. Deletes interactions, messages, marketing, carts, tickets, appointments, behavioural data; anonymizes the profile in place; disables and scrubs the login at the IdP. Bills/payments/orders/usage/agreements are retained with their legal basis named in the report. A partial fan-out answers `"status": "partial – re-run required"` — re-run until `completed`. `GET /privacy/v1/erasure` lists the audit trail.

## Retention clocks

`BSS_RETENTION_INTERACTIONS_SECONDS` (party-interaction) and `BSS_RETENTION_TICKETS_SECONDS` (trouble-ticket, settled tickets only) — 0 = keep forever (default). Production sets day-scale values (e.g. 63072000 = 2 years); `BSS_RETENTION_SWEEP_MS` paces the sweep. TickGuard-guarded: replicas never double-sweep.

## PCI & lawful intercept, honestly

Card data never enters the BSS — the vault stores `pspToken/brand/lastFour/expiry` (SAQ-A-shaped; a QSA attests the rest). Lawful intercept: the BSS half is warrant-gated subscriber disclosure — shaped, not built; see [docs/privacy.md](https://docs.genalpha-bss.com/privacy.md).

## 15 • The agent channel (agentic commerce) — suite #64

**Being shopped by AI agents is opt-in.** Every operator carries `agent-commerce: off | discovery | full` in the tenant registry — `off` (dark to agents, 404), `discovery` (the product feed answers; agent checkout refused with 403 — findable, checkout stays human), `full` (delegated checkout too). Newborn operators are born `off`. Flip it on the operator form (console → Operators → Agent commerce) or by env ( `AGENT_COMMERCE` , `AGENT_COMMERCE_<ID>` ); the gateway enforces it live — no restart.

### The surface (ACP, spec 2026-04-17)

**Feed:** `GET /acp/product_feed` — public projection of the active TMF620 catalog (unconditioned prices only, one-time preferred; unpriced offerings are skipped). **Checkout:** `POST /acp/checkout_sessions` with `{items:[{id,quantity}]}`, then `GET/POST /acp/checkout_sessions/{id}`, `POST .../{id}/complete` (needs `Authorization` + `Idempotency-Key` + `payment_data.token`), `POST .../{id}/cancel`. A session rides a real TMF663 cart; complete charges the payment token and places a TMF622 order marked `category=agenticCommerce` with the buyer as relatedParty. A replayed `Idempotency-Key` returns the *same* order; a fresh key on a completed session is 409.

### Delegated checkout (the trust model)

Agents never hold a machine identity. The shopper's token is exchanged (OAuth token exchange, Keycloak ≥26.2) through the confidential `bss-agent` client — `fullScopeAllowed` `off`, scope-mapped to exactly `catalog:read` `ordering:read/write` `payment:read/write` — so the agent's credential can order and pay and nothing else. The shopper's client must carry an audience

mapper naming `bss-agent` : the realm grants the delegation path explicitly, or the exchange is refused. Suite #64 decodes both tokens to prove the downscoping.

## MCP tools (Claude-class agents)

`integrations/mcp-server` wears the same surface as five retail tools: `search_offerings` , `get_offering` (with "also bought"), `start_checkout` , `update_checkout` , `complete_checkout` . Shopper env: `BSS_SHOPPER_USERNAME/PASSWORD` ; exchange client: `BSS_AGENT_CLIENT_ID/SECRET` .

Honest scope: the merchant surface is implemented and proven. Appearing *inside* ChatGPT/Perplexity additionally requires the operator's business onboarding with those platforms (feed registration, PSP delegated-payment agreement) — an operator act, not a code gap. The endpoints they register are the ones suite #64 exercises.

## 16 · The digital workforce — suite #65

**The badge is the opt-in.** An AI worker is hired like staff: mint a login, grant the `digital-worker` bundle (console → Staff, or `integrations/hermes-worker/hire-worker.sh` ). The grant sheds the walk-in customer defaults — a worker is staff. Revoke the badge to fire; the tenant's AI kill-switch stops the whole workforce mid-shift.

### The job

`GET /ai/v1/workforce/tasks` — the open queue, derived LIVE from real backlogs (unassigned TMF621 tickets, unapplied cash). `POST .../tasks/{id}/claim` (15-min lease, `BSS_WORKFORCE_LEASE_SECONDS` ), `.../complete` (VERIFIED: a ticket task completes only when the ticket is actually resolved), `.../escalate` (a reason required — escalations are counted, not punished). `GET .../ledger` is the shift record, including the worker's self-reported model usage (labeled: its own word — the worker brings its own LLM).

### Approvals (the T3 gate)

Refunds, cease, erasure: the worker FILES ( `POST .../approvals` — method + `/tmf-api/` path + body + reason); nothing executes. A human approves on the Workforce tab and the stored action runs under *the approver's own token*; refusing keeps its receipt too. Filing is `workforce:use` ; deciding is `ai:use` — a worker can never approve its own ask.

### The scoreboard — and the controls



Console → **Workforce**: completed / escalated / deflection, average handle time, **reopen rate** (completed tickets re-checked live — the honesty metric), the crew by name and observed type, approvals queue, shift ledger, self-reported cost (labeled), and human-minutes saved — an estimate that SAYS so, computed from operator-set baselines ( `BSS_WORKFORCE_BASELINE_MINUTES_TICKET/-CASH` ). The tab also holds the operator's two levers: **Crew ceiling** (governance `maxWorkers` ; a new worker past it is refused at claim time — surge staffing never outgrows it) and **Hire a worker** (pick a job — care / back-office / generalist — the badge is minted and the credentials shown ONCE with the matching job card to run; activating is the runtime's act, firing is the Staff-tab revoke).

## Surge staffing

The staffing signal — `staffing` in the KPIs, Prometheus gauges ( `bss_workforce_backlog_depth / _active_workers / _surge` , per tenant) and `WorkforceSurgeEvent` on the boundary — tells an autoscaler when the crew is behind. Scale with KEDA ( `integrations/hermes-worker/k8s-surge-example.yaml` ) or the dev watcher ( `surge.sh` ); the ceiling caps whatever the scaler does. The BSS signals and refuses; it never spawns worker processes itself.

## The Hermes package (day 1)

`integrations/hermes-worker/` : `hire-worker.sh` (the badge, shown once), `config.snippet.yaml` (Hermes `~/hermes/config.yaml` MCP block with the tool whitelist), and `SKILL.md` job cards (care-triage every 15 min, cash-matching daily). Runtime-neutral by construction — any MCP-speaking agent hires into the same job, badge and audit.

# 17 • The wrapped legacy estate (overlay) — suite #67

**A legacy BSS is just another seam.** Three per-tenant registry fields — `legacy-catalog-base-url` , `legacy-fulfilment-base-url` , `legacy-ticket-base-url` (empty = native mode) — wrap an existing estate: its catalog federates read-through into the native list and the ACP feed (legacy-prefixed, cached, fail-soft); orders carrying `legacy-` items hand off to the legacy work-order queue; the digital workforce works the legacy incident backlog age-stamped, with completion VERIFIED against the legacy system's own state. Never two writers: legacy masters its data, genalpha masters engagement. Also: the workforce package's closed loop — dashboard Hire starts a running worker container via the controller ( `--profile workforce` ); the worker's LLM is live registry config ( `worker-ai-*` ) rolled onto by the controller.

## 18 · Generative discoverability (GEO) — suite #68

**AI crawlers read HTML only — and the operator holds the switch.** Per-tenant ai-visibility: open | search-only | dark executes at robots.txt: open welcomes all crawlers (+ llms.txt, honestly labeled speculative); search-only keeps classic search and disallows AI answer/training bots (GPTBot, ClaudeBot, PerplexityBot, Google-Extended, CCBot...); dark disallows everyone. Newborn operators default search-only .

The gateway DUAL-SERVES /shop/offering/{id} by User-Agent: crawlers get bot-readable HTML with schema.org Product/Offer JSON-LD rendered live from TMF620 (no authoring, no syncing — suite #68 proves bot price == catalog price); humans get the SPA unchanged.

/robots.txt , /sitemap.xml , /llms.txt generate per tenant from the same source. Follow-ups, named: insight ai-answer referrer tagging; knowledge-base FAQ pages as a public help center.

## 19 · The capability map

**What's here, what isn't** — TM Forum business capabilities (verb-first, implementation-free) mapped to tech capabilities (ODA functional blocks / TMF Open APIs) and the component carrying each, across eTOM-style domains. Every  cites its proof suite; every  carries its caveat (mock-default PSP, OCS facade, NRDB shaped-not-connected); the  column names what a deployment brings from elsewhere (production OCS, tax engine, ERP/GL, roaming settlement, fraud/RA, FSM, real OSS) — each behind an existing seam or the overlay pattern (suite #67). The full map: <docs/capability-map.md> · executive view (filterable, GENERATED from the md so the faces cannot drift): <capability-map.html>.

## 20 · Loyalty — reward & retain — suite #69

**The program is data; membership is a choice.** Component #34 ( loyalty , TMF658, :8114) is a points LEDGER wired between engines that already exist. The program — earn rate, gigabyte price, voucher percent, tier thresholds, expiry clock (0 = never) — is marketer-editable data (campaign:write). Customers OPT IN on the storefront My page or the app's My-points card; nobody is enrolled by default. Every point movement is an append-only journal row carrying its cause: bill:<id> , redeem:... , expiry:<date> , adjust:<reason> (goodwill credits REQUIRE a reason).

**Earn** — a settled bill earns at the program's rate, idempotent per bill (the journal's unique cause index), members only. **Burn** — three ways, each landing on an existing engine and verified there,

never on loyalty's own word: points → gigabytes on THIS month's meter via the usage allowance-boost mechanic; points → a single-use discount voucher minted as a REAL TMF671 promotion (unique code, valid at the same `checkPromotion` door as any campaign code, second redemption 409s); tiers → PRICING, because `loyaltyTier` rides the billing pricing context and one console-authored rule ("Loyalty tier benefit" preset) prices gold differently — the pricing engine needed zero new code. **Govern** — points are a liability: the TickGuard-guarded expiry sweep journals every expired point (FIFO arithmetic), and `GET .../liability` returns the outstanding-points number a finance team books.

**Retention wiring** — `LoyaltyTierChangedEvent` rides the outbox onto `bss.loyalty.events`, which the campaign engine hears like any business event: the console's campaign form now opens with RECIPES ("Churn save: a loyalty offer to at-risk customers" on the churn scorer's alert; "Loyalty: congratulate a tier change") that prefill trigger + message and stay fully editable. Suite #69 proves all nine legs end to end, including a real tier flip landing a real TMF681 message, and paula's points card composed onto her app Home.

## 21 • Revenue export — the subledger — suite #70

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**The BSS is the subledger; the ERP is the ledger.** Component #35 ( `revenue`, house API `/revenue/v1`, :8107 — no TM Forum Open API covers the GL, deliberately) turns billing and payment events into balanced double-entry journal postings: invoice issued → AR debit / revenue credits per rate line (discounts as contra-revenue); cash books ONCE, at payment capture, whichever door the money came through (card, CSR, bank giro via remittance); refunds book the reverse. Balance is a save-time INVARIANT and every posting is idempotent per source ref — at-least-once delivery can never double-book.

**The chart is data** — `GET/POST /revenue/v1/accountMapping` (billing:admin): finance's own codes and names per posting key; booked lines keep the snapshot they were born with, so a remap never rewrites history. **The export** — `GET /revenue/v1/journalExport?date=` (CSV, one row per line) is the file a period-close import ingests; `journalEntry` serves the same data as JSON; the console's read-only **Journal** tab shows entries with their lines. **The tie-out** — `GET /revenue/v1/reconciliation?date=` : per-account totals, billed vs cash, every entry balance-checked live, and the loyalty points liability as a labeled CONTROL number (no currency valuation is invented — pricing a point is P2 config). `POST /revenue/v1/backfill` onboards pre-arc bills idempotently. P2 dials, all data: a VAT percent on the `tax` key splits postings into net revenue + tax payable (gross-minus-nets — rounding can never unbalance); a per-point value on `loyalty:liability` turns the points control number into a daily booked accrual; `POST /periodClose` makes the export FINAL (postings into a closed period 409); dispute credits on unpaid bills book contra-revenue automatically; `?format=sap|netsuite` emits ERP-shaped

layouts. Rev-rec (ASC 606/IFRS 15) stays in the ERP's engine; this feed is its clean input. See [docs/revenue-export-plan.md](#).

## 22 • Credit notes — suite #71

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**The wrong invoice is never edited** — a numbered credit note reverses it. `POST /customerBill/{id}/creditNote {amount?, reason}` (billing:admin — agents open disputes, admins decide money; the 403 wall is suite-proven): mints `CN-NNNNNN` from billing's gapless per-tenant `document_sequence` (row-locked — the bookkeeping rules demand an unbroken series), references the original bill number, and **REQUIRES** a reason. Unpaid bill → the due comes down (a negative `creditNote` line says why; zero settles). Settled bill → the money moves **BACK** through the PSP (`settlement refunded`, `refundRef` kept). Dispute resolutions with `outcome=credit` now mint their own credit note (`disputeId` carried). Customers read their own notes (`GET /creditNote`, PDF at `/creditNote/{id}/document.pdf`) — the storefront shows a `kreditnota` chip per bill; the CSR 360 has the issue button. The revenue subledger books **REDUCED** notes as contra-revenue under the document's number and books nothing extra for **REFUNDED** ones (the refund event owns that). P2 shipped: `GET /creditNote/{id}/document.xml` is the UBL `CreditNote-2` (type 381, EHF/Peppol shape, `BillingReference` to the invoice) and partner-wired tenants ship it on the **SAME** distribution ledger as bills; `lines:[{id, amount?}]` credits named rate lines (each reversal names what it reverses, capped per line); the back-office console's Disputes tab is the decide worklist.

## 23 • Process flows & the incident agent — suite #72

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**Choreography runs the flows; TMF701 explains them.** Component #36 (`process`, :8116, `/tmf-api/processFlowManagement/v4`): design intent as editable `processFlowSpecification`s (each task's owed event + time allowance), run-time `processFlow`s **PROJECTED** from ordering/SOM events with the per-flow cross-system timeline persisted, and **STUCK AS A STATE** — the TickGuard sweep fails a task past its allowance, out loud on `bss.process.events`. Operators recover via `PATCH .../taskFlow/{id}`; customers see their own flows (self-service order tracking); console: Process flows tab.

**The agent whose memory compounds** (in `intelligence`, IG1547-aligned): a failed task triggers context assembly (spec + failed step + timeline) and a **GOVERNED** diagnosis — same kill-switch/budget/meter/audit doors as every AI call; its **ONLY** write is a ticket note (LO). Every investigation is an episodic trace with a **MANDATORY** human verdict (`POST /ai/v1/incident/{id}/verdict`). Three useful verdicts on one signature draft a runbook (governed, provenance to source traces) that stays `proposed` until approved (console: Runbooks

tab; deciding — approve/reject/revoke — is `ai:admin` like contracts and budgets, `ai:use` reads the library); the next occurrence auto-diagnoses FROM MEMORY — no model call, audit as witness — and `GET /ai/v1/incident/stats` carries the learning curve as data. Revocation sends diagnosis back to the LLM path: the brake is first-class. L2 auto-remediation stays out of scope until per-taskFlow idempotency is earned. See [docs/process-memory-plan.md](#).

## 24 · The standards sweep — suites #73–76

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**Fulfilment (TMF700+697, component #37, :8117)** — physical orders mint `shippingOrder s` (warehouse PATCHes states, tracking ref), install bookings mint `workOrder s`; DELIVERED + visit COMPLETED completes the product order machine-driven. Customers track their own deliveries; state changes are staff-only. **SLA (TMF623)** — an `sla` block on the agreement characteristic (circuit, thresholdMinutes, creditAmount, capPerMonth); late-resolved problems mint violations on assurance's capped ledger ( `/tmf-api/slaManagement/v4/sla|slaViolation` ) and billing compensates with the PRE-AGREED credit note, in-process, no one deciding at breach time. **Event hub (TMF688, component #38, :8123)** — `POST /hub` callback + filter (billing:admin), delivery ledger with backoff retries and dead-letter. **Thin faces:** TMF724 incidents over the agent's memory (verdicts are the state machine), TMF653 serviceTests with history over diagnose, TMF639 pools + issued-number ledger with the honest generator note. See [docs/standards-completion-plan.md](#).

## 25 · The product configurator (TMF760) — suites #77/#78

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Two anonymous task resources on product-catalog at `/tmf-api/productConfigurationManagement/v5` (TMF760 has no v4), stateless — the POST computes from the catalog's own rows and answers. **Query:** `POST /queryProductConfiguration` with `{productConfiguration:{productOffering:{id}}}` returns the configuration space — fixed members, choice groups ( `minSelections` / `maxSelections` / default), each option's pickers and prices with `appliesWhen` conditions visible. **Check:** `POST /checkProductConfiguration` with `selectedOption:[{id}]` + `configurationCharacteristic:[{name,value}]` validates BOTH pick bounds, characteristic values scoped to the PICKED options, and policy block rules (under `bss-catalog` , the catalog's machine identity; fail-open) — rejections use ordering's exact wording. An approved answer carries `configurationPrice` (monthly + one-time + lines; conditioned lines only on exact match; the deal engine's indicative total rides along) and an ORDER-READY `productConfiguration` echo with each pick on the option whose spec owns it. **Agent channel:** MCP tools `get_configuration_options` and `configure_product` (an approved answer includes a ready `checkoutItem` ); ACP checkout items accept `configuration` — the cart prices from the check, refuses invalid picks before a session opens, and complete



builds nested order items with `product.productCharacteristic`, so billing rates the configuration like any other channel's order. See [docs/tmf760-configurator-plan.md](#).

## 26 · Service qualification & the coverage map (TMF645) — suites #79/#80

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TMF679 answers the commercial question (may this offering be sold here); TMF645 answers the technical one, on the same `qualification` service at `/tmf-api/serviceQualificationManagement/v4`. The substance is the **coverage map** — operator-editable rows `{technology, postcodePrefix, maxDownMbps, maxUpMbps}`; an EMPTY prefix matches everywhere (the mobile fallback), longest matching prefix wins per technology; seeded by `ops/seed/seed_coverage_map.py`, CRUD gated by `qualification:write`. **Query:** `POST /queryServiceQualification` with `{searchCriteria:{place:{postCode}}}` — every technology deliverable there, with bandwidth characteristics. **Check:** `POST /checkServiceQualification` — verdict per requested service (name the technology via `serviceSpecification.name` or a characteristic; optional `minDownstreamMbps`), and NEVER a bare no: a refusal carries `eligibilityUnavailabilityReason` AND `alternateServiceProposal` with the best technology the address can have. Checks persist and read back anonymously by unguessable id; the LIST (customer addresses) is back-office. **Channels:** ordering qualifies every placed item at create (same TMF679 check the storefront runs; fail-open; refusals speak the qualification's words), and the cart's address block names what the network delivers ("Fiber up to 1000 Mbit/s at your address"). See [docs/tmf645-service-qualification-plan.md](#).

## 27 · AI management — the control plane's face (TMF915) — suite #81

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The governed-AI machinery, standards-addressable on intelligence at `/tmf-api/aiManagement/v4`. **GET /aiModel** (ai:use): the models the audit ledger PROVES have served — distinct provider/model/tier pairs with the scenarios each served, plus the trained churn classifier carrying its `trainingRecord` (`sampleCount`, `positives`, `trainedAt`). **GET /aiModelContract/{id}** (ai:use): one contract per governed scenario — `state` (`active` | `suspended` | `haltedByKillSwitch`), `servedBy` model refs, `monitoring` (`calls`, `tokens`, `costMicros`, `avgLatencyMs`, and per-outcome counts including `refused-budget` / `refused-disabled` / `refused-contract`), and the `guardrail` block (`tenant kill-switch`, `budget`, `window`). **PATCH /aiModelContract/{id}** (ai:admin — the seam, now open): `{state: suspended|active, note}` — the per-scenario brake, checked in the governor's gate order right

after the tenant kill-switch; a suspended contract's calls 403 and land on the ledger as `refused-contract`. Budget-setting ( `POST /ai/v1/governance/budget` ) now also requires `ai:admin` — as do runbook decisions and workforce hire/fire (the worker-controller probes `GET /ai/v1/governance/adminCheck` and trusts the BSS's 200/403 as the verdict; the console hides `ceiling+hire` from non-admins, who keep the scoreboard and approvals). Ops note: `accessTokenLifespan` is 1800s in BOTH realm files — a Keycloak container recreate re-imports realms from files, so live-only settings (and dynamically-onboarded realms like aurora) do not survive one. See <docs/tmf915-ai-management-plan.md>.

## 28 · Risk management — the door check (TMF696) — suite #82

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The acquisition-side twin of churn, on intelligence at `/tmf-api/riskManagement/v4` behind `risk:assess` (demo staff + the bss-ordering machine; customers 403 even on their own scores — a risk score is a credit check). **POST /partyRiskAssessment** `{relatedParty:[{id}]}` and **POST /productOrderRiskAssessment** (adds `totalQuantity`, `lineCount`, `verifiedIdentity` — supplied by the caller whose session knows it; a verified session scores 20 lower). The score is TRANSPARENT: named signals (`unpaidBills`, `creditNotes`, `orderVelocity`, `newAccount`, `bulkOrder`, `verifiedSession`) each with points + raw evidence in the body — recomputable by hand. Deliberately NOT scored: failed payments (refusals never persist) and lifetime verification (session claim only). Assessments persist and read back by id. **Enforcement is data:** ordering fetches a fresh assessment (machine identity, fail-open) into the policy context as `riskScore` / `riskLevel`; the operator's JsonLogic rule sets the threshold — deny returns 422 `POLICY_DENIED`. Billing's `GET /creditNote` grew a `relatedPartyId` filter for the engine. See <docs/tmf696-risk-plan.md>.

## 29 · Partnership types & geographic sites (TMF668 + TMF674) — suites #83/#84

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**TMF668** on the agreement service at `/tmf-api/partnershipTypeManagement/v4/partnershipType` (reads `agreement:read`, writes `agreement:write`): a partnership KIND is `{name, description, status, roleType:[{name, description}]}` — a kind without roles is refused. An agreement with `agreementType: "partnership"` whose `characteristic.partnershipTypeId` names a kind is VALIDATED at create: every `engagedParty.role` must be one the kind permits (refusals name the permitted list; unknown/retired kinds refuse outright). Untyped agreements are untouched. **TMF674** on geographic-address at `/tmf-api/geographicSiteManagement/v4/geographicSite` (reads

`address:read , writes address:write )`: a site is `{name, status: planned|active|retired, relatedParty, place:{id}}` where `place.id` MUST reference a stored TMF673 `geographicAddress` (validated at create, embedded on every read); filter with `?relatedPartyId=` ; PATCH moves the lifecycle; a site's address is reusable on orders — entered once. See [docs/tmf668-674-thin-tail-plan.md](https://docs.tmf668-674-thin-tail-plan.md).

## 30 · The proof run & fleet-age runbooks — suite #85 and the receipt

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**The proof run:** `ops/run-all-suites.sh` runs all suites serially with a readiness gate (IdP + gateway must answer before each suite), inter-suite pacing (a young fleet out-runs its own per-subject rate limiter), the surge controller parked except for the suite that tests it, onboarding suites first (a recreated Keycloak forgets dynamic realms), rate-limit suites last, and one automatic retry pass with attempts recorded. The receipt renders to `docs/proof-run.html` via `docs/build-proof-run-html.py`. **The pruning runbook:** `ops/prune-dev-fleet.sh` retires active subscriptions owned by anyone who is not a named demo persona (nothing deleted — products stop billing), then seed one usage record per persona (meters derive from monthly usage). **Fleet-age laws** (from [docs/proof-run-findings.md](https://docs/proof-run-findings.md)): machine list-clients page to exhaustion or filter server-side (an unfiltered list ages out of any fixed page — `GET /revenue/v1/journalEntry?sourceRef=` exists because of this); batch endpoints get batch timeouts; suites probing aged personas should mint purpose-made customers; ad-hoc suite runs park `bss-worker-controller` first. Billing runs on a worker pool (`bss.billing.run-concurrency` , default 8; crash-resume proven by suite #57 against the pool) — suite #85 asserts the aged-fleet run completes inside suite patience.

## 31 · The shop that knows you (personalization) — suite #86

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**Usage-aware upsell:** the signed-in `GET /ai/v1/forYou` payload carries an `upsell` block when any meter is  $\geq 80\%$  used — `{bucketName, usedPct, currentAllowance, suggestedOffering, suggestedAllowance}` , the suggestion being the NEXT rung of the allowance ladder (the usage service's `usageAllowance` rows per offering, smallest larger allowance for the same usage type; rows without an offering name get a descriptive label). No meter, a comfortable meter, or no fuller rung → no block. The Shop page renders it as `data-testid=upsell-card` above the recommended rail. **Returning-visitor rules:** the personalization rule context now carries `visits` (distinct event-days per visitor) — author `{">=": [{"var":"visits"}, 2]}` rules in Rules or via the copilot. **Full-profile grid:** the Shop



grid ranks singles by the visitor's whole consented `interests[]` array in browse-weight order; a rule's `heroCategory` still floats to the top. Pricing on the grid is deliberately NOT a personalization lever. Hygiene note: killed suite runs orphan their finally-cleanup policy rules — if default experiences misbehave, list `domain=personalization` rules for leftovers. See [docs/personalization-shop-plan.md](https://docs.personalization-shop-plan.md).

## 32 · The CTK campaign — twenty-five kits, the closed list

**Scorecard:** [docs/ctk-conformance.md](https://docs/ctk-conformance.md) — twenty-five official CTKs at zero failures; TMF676/681 stay intentional partials. **The list is CLOSED:** the `tmforum-rand` org publishes no kit for the remaining faces (688/696/760/915/700/697/701/724/623) — there is nothing left to run.

**Runners:** modern kits (`config.json` + `index.js`) via `ops/ctk/runctk.py` ; R18-era kits (raw collection + environment) via `ops/ctk/runctk-r18.py` — both structure URLs for modern newman, bake the bearer, run with `--timeout-script 120000` . Gotchas that will bite again: pass the kit dir by **absolute path** (relative once made newman fail silently and the summary read a STALE `result.json`); some kits write `pmtest.json` BEFORE their own payload injection (`runctk.py` injects config payloads itself now); secondary host-vars ( `{{Service_Ordering}}` )/tmf-api/... ) hang newman on literal-var DNS — `runctk-r18.py` resolves them to the gateway origin.

**One-time kit prep** (documented in [ops/ctk/README.md](https://ops/ctk/README.md)): TMF674's example payload must reference a STORED TMF673 address; TMF638 assumes a sandbox — seed two uniquely-named probe services, the newer in state `designed` . **Contract changes the campaign shipped,**

**fleet-wide:** `agreement type` (= `agreementType` ) and `ticket ticketType` are now MANDATORY (storefront files `support` , self-heal files `incident` ); a roleless partnership kind stores but permits NOTHING at signature; `serviceTest` requires `testSpecification` (the built-in virtual spec is `diagnose` ); `agreement` + `ticket` lists page NEWEST-FIRST. New faces: read-only TMF633 `serviceSpecification` on SOM (so inventory spec refs dereference), northbound `POST /serviceOrder` (external orders recorded `acknowledged` ), `POST /serviceProblem` (declared problems), and v3 task dialects for TMF645/679 that delegate to the v4 coverage/serviceable-area engines. **Gateway:** SCG 4.2+ only adds `X-Forwarded-*` for `trusted-proxies` (now set to the private ranges); services build absolute `Location` headers from those — keep that setting when changing gateway config. **Workforce suites:**

`workforce_runtime / agentic_workforce` need `bss-worker-controller` UP (start for them, stop after, then `docker rm -f wf-*` ).

genalpha-bss · eighty-six suites, twenty-five CTKs at zero failures · the build story is *The Honest Machine* (docs/book) · the TMF vocabulary is docs/book/tmf-reference.md